

Supporting Information for

Uncovering thousands of endosymbiont DNA transfer events within single cockroach genomes

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Supporting Information Text

DNA and RNA extraction, library preparation, and sequencing

This study involved sequencing and assembling genomes from 8 Blaberidae species (Dataset S1), analysed with previously generated genomes from 15 Blattodea and 77 *Blattabacterium* (Dataset S2). Specimens of *Macropanesthia lithgowae*, *Neogeoscapheus dahmsi*, *Panesthia lata*, *Panesthia sloanei*, *Panesthia tryoni tegminifera*, *Panesthia tryoni tryoni* (Eungella), *Panesthia tryoni tryoni* (Kroombit Tops) and *Parapanesthia gigantea* were collected throughout Australia. Genomic DNA was prepared from the leg muscle tissue of *M. lithgowae*, *P. tryoni tryoni* (Kroombit), *P. sloanei*, and *P. lata*, while the DNA of *Par. gigantea*, *P. tryoni tegminifera*, *P. tryoni tryoni* (Eungella) and *N. dahmsi* was extracted from both leg muscle and fat body tissue. Live cockroaches were chilled at -30°C for 4–5 min to induce torpor, and the femoral muscle (~3 mm) was dissected from one or two hind legs, depending on body size. Following this, insects were frozen at -80°C. Tissue was incubated overnight at 56°C in 600 µL SDS buffer with 20 µL of 20 mg/mL stock Proteinase K (final concentration 400 µg/mL), followed by RNase A treatment (final concentration 20 µg/mL) at 37°C for 1 h.

DNA extractions were performed using the phenol-chloroform method or using the NEB Monarch HMW DNA Extraction Kit (NEB #T3060). The phenol-chloroform method involved two extractions with an equal volume of phenol:chloroform:isoamyl alcohol (25:24:1), isopropanol precipitation with sodium acetate (3 M, pH 5.5). The pellet was washed twice with 70% ethanol, air-dried, and the DNA pellet resuspended in 40 µL of 10 mM Tris (pH 8.0). The NEB Monarch HMW DNA Extraction Kit was used according to the manufacturer's instructions, with the exception that Proteinase K digestion was undertaken for 4 h at 600 rpm, rather than the recommended 45 min, to ensure thorough digestion of the tissue. As minimal gDNA appeared to stick to the glass beads, the supernatant was centrifuged for 10 min at 16,000 g at room temperature, the pellet was washed twice in 500 µL kit Wash Buffer, dried briefly, then resuspended in 20 µL Elution Buffer II. The quality and integrity of the gDNA was assessed by a 4200 TapeStation System (Agilent Technologies) using Genomic DNA ScreenTape (Cat No. 5067-5365) and reagents (Cat No. 5067-5366). The absorbance ratios were assessed by Nanodrop Spectrophotometer (DeNovix DS-11). Concentration was determined using a Qubit 2.0 Fluorometer (Invitrogen by Life Technologies) using a dsDNA BR assay kit (Cat No. Q32850).

Long-read sequencing was performed using gDNA from *M. lithgowae*, *N. dahmsi*, *P. lata*, *P. sloanei*, *P. tryoni tryoni* (two localities, from different clades), *P. tryoni tegminifera* and *Par. gigantea* (Dataset S1). Libraries were prepared from gDNA using the Ligation Sequencing Kit V14 (SQK-LSK114) protocol according to the manufacturer's instructions with the following modifications and options. The DNA Repair and End-Prep reaction continued for 15–20 min. The Adapter ligation reaction proceeded for 20 min and the kit LFB wash buffer was used to enrich for longer fragments. Final library elution occurred at 37°C to improve the recovery of long fragments. The integrity and concentration of final libraries were assessed using a 4200 TapeStation System (Agilent Technologies) and Qubit 2.0 Fluorometer (Invitrogen by Life Technologies) as previously described.

Libraries were sequenced on a PromethION 24 sequencer (Oxford Nanopore Technologies) using R10.4.1 flow cells (FLO-PRO114M), following the manufacturer's instructions. Libraries were stored at 4°C for less than a week if not sequenced immediately. The sequencing output was additionally increased by washing the flow cells (EXP-WSH004) after approximately 24 and 48 h, and reloading the libraries for sequencing for a total of 72 h. Sequencing was performed in super-accurate base-calling v4.3.0 mode (400 bps).

RNA was extracted from leg and fat body tissue samples from *Panesthia tryoni tegminifera* (Z011), *Macropanesthia lithgowae* (Z009) and *Panesthia sloanei* (Z001) using Trizol, following the manufacturer's recommendations. RNA samples were submitted to the Australian Genome Research Facility (AGRF) for library preparation and sequencing. RNA quantity and integrity were assessed by AGRF prior to library construction. Sequencing libraries were prepared using an

Illumina-compatible stranded total RNA library preparation protocol incorporating ribosomal RNA depletion. Following rRNA removal, RNA was fragmented and reverse-transcribed to cDNA, followed by end repair, adaptor ligation, and PCR amplification. Libraries were pooled and sequenced on an Illumina platform to generate paired-end reads. Base calling, demultiplexing, and primary quality control were performed by AGRF prior to delivery of FASTQ files.

Genome assembly and annotation

For the ONT sequencing that incorporated duplex base calling (Dataset S1), redundant simplex reads (i.e., simplex reads which have duplex offsprings) were removed using samtools v1.2 [1]. Adapters were trimmed for all ONT sequence data (with and without duplex calling) and converted into fastq files using dorado v0.6 (<https://github.com/nanoporetech/dorado>).

De novo assembly was carried out using the trimmed ONT data using Flye v 2.9.3 [2]. After some initial optimization experiments, the ONT sequencing data with duplex calling was assembled using the 'ONT corrected' mode (--nano-corr) and a minimum overlap of 10,000 (--min-overlap 10000), while the other ONT sequencing data was assembled using the 'ONT raw' mode (--nano-raw) and default settings. An additional two iterations of polishing were carried out after the assemblies. The purge_dups v1.2.5 [3] pipeline was used to remove putative haplotypic duplications and heterozygous overlaps from the genomes. Raw ONT reads were mapped to the draft assembly using minimap2 v2.18 [4,5] for this purging step. QUAST v5.3.0 [6] was used to assess the assembly continuity, and BUSCO v5.8.1 was used to assess assembly completeness (utilizing the Insecta lineage data set; [7,8]).

The genome assemblies were annotated using FGENESH++ v7.2.2 [9,10], performed on the Nimbus cloud service provided by the Pawsey Supercomputing Centre. Prior to annotation, we generated a repeat library with RepeatModeler v2.0.1 [11] and masked repetitive elements in the assemblies using RepeatMasker v4.0.6 [12], applying the '-nolow' option to exclude low-complexity or simple repeats from masking. The masked and unmasked assemblies were used as input for FGENESH++. Following Ewart et al. [13], FGENESH++ was run with the non-mammalian general pipeline settings, employing optimized gene-finding parameters trained on the pea aphid (*Acyrtosiphon pisum*) genome, and a curated insect protein database (provided by Softberry; 2020 release) for "prot_map" homology-based predictions. *Ab initio* predictions lacking BLAST support were excluded.

These newly assembled genomes were combined with other cockroach and termite genomes from InsectBase 2.0 [14], NCBI and Ewart et al. [13] for subsequent analyses (see Dataset S2 for species/genome details). *Periplaneta americana*, *Blattella germanica*, and *Zootermopsis nevadensis* had multiple available genomes from various sources, which were all included to assess whether differences in data type and/or assembly methods affected the results of subsequent analyses.

RNA assembly and mapping

Raw RNA reads underwent secondary quality assessment using FastQC v0.12.0 [15], followed by adaptor trimming and removal of reads with length <20 bp and Q-scores <20 using the BBDuk function in *BBTools* v.39.62 [16]. We then mapped the RNA to their corresponding genomes using *HISAT* v.2.2.1 [17] with default alignment scores and mapping stringency, before assembling the mapped reads into contigs using *StringTie* v.3.0.2 [18]. Candidate coding regions were identified with *TransDecoder* v.5.7.1 [19]. We used a custom awk script to remove duplicated and incomplete transcripts, retaining only the longest, complete transcript per gene (see relevant scripts on GitHub: https://github.com/MEEP-projects/HGT_pipeline).

To determine whether HGT sequences were transcribed, we used blastn v2.2.3 to align the fasta files containing unique inserts (i.e. filtered for duplication and transposition) against the finalized transcript files. Output BLAST results were processed using a custom R script, whereby the coordinates of the BLAST hits were compared against the corresponding genome annotation files to identify whether they represented inter- or intra-genic regions, and in turn exonic or intronic

regions. For inserts inferred to coincide with exonic regions, we investigated the potential functions of the associated genes by examining their homology with previously annotated genes in FGENESH++ (Dataset S4).

Characterizing HGT inserts

The following steps were carried out to identify, filter, and date the inserts from *Blattabacterium* in the cockroach and termite genomes. This pipeline workflow and the relevant scripts are available on GitHub (https://github.com/MEEP-projects/HGT_pipeline). Alternative thresholds were applied for the several filtering steps, specified in brackets below. The corresponding results when applying these alternative thresholds are presented in the 'HGT insert QC' section and Dataset S3 (also see the 'HGT insert QC' section).

We first created a library comprising 77 *Blattabacterium* genomes (see Dataset S2 for genome details), each of which was broken into 150 bp fragments (alternative threshold = 100 bp) using a custom AWK script. These *Blattabacterium* 'fragments' were aligned to the reference host genome using bwa mem v0.7.17 [20], implementing default parameters. The resultant alignment was converted into a bam file with samtools, then into a bed file using bedtools v2.3 [21]. Regions of the cockroach genome against which *Blattabacterium* fragments aligned were identified as potential horizontal inserts. In the case that aligned *Blattabacterium* fragments overlapped on a host contig, or aligned fragments were found within 150 bp of each other on a host contig (alternative threshold = 50 bp), we merged the host region aligning to these sequences as a single putative HGT insert. Putative inserts shorter than 50 bp (alternative threshold = 30 bp) were removed using AWK. Note that inserts were detected in genomes of species for which only leg muscle tissue was included during DNA extraction (thus excluding *Blattabacterium*), and also species for which fat body was included (thus including *Blattabacterium* cells and potentially genomic contigs).

Several filtering steps were carried out to remove potential artifactual HGT inserts. First, we conservatively removed any contigs in which genuine *Blattabacterium* sequences may have been combined with cockroach sequences during genome assembly (and therefore erroneously characterized as HGT inserts). We removed any contigs for which the total length of putative bacterial sequences (identified using the method above) was $\geq 20\%$ of the total contig length. In addition, we conservatively removed any contigs with one or more contiguous putative HGT inserts above 10,000 bp in length (which were found to be significant outliers based on the length distributions of all putative inserts in our dataset; see Fig. 1B). Second, inserts that formed part of false assembly duplications were identified and removed. To identify these potential false duplicate inserts, the insert regions were extracted from the genomes with 300 bp of flanking sequence (to either side) using bedtools, then BLASTed against all of the other inserts (with flanks) within the same genome using blastn v2.2.3. Inserts were considered false duplicates and removed if the resultant sequence identity was $>90\%$ (alternative threshold $>95\%$), and query coverage was $>90\%$ (alternative threshold $>95\%$). Third, highly repetitive low-complexity ("low-entropy") sequences were identified using the symmetrical DUST algorithm [22] implemented in minimap2, and any inserts with contiguous, low-entropy regions representing $\geq 50\%$ of the total insert length were removed. Fourth, to ensure that we could characterize the cockroach genomic sequence surrounding each insert, we excluded inserts where either of the 300 bp flanks consisted of $\geq 50\%$ ambiguous (N) nucleotides. Lastly, we removed inserts that were more similar to other bacterial species (i.e., not *Blattabacterium*), as they could be potential artifacts or genuine inserts of other related Bacteroidales. Each insert was BLASTed against two custom databases: one comprising only *Blattabacterium* sequences ($n = 77$; Dataset S2), and one with other bacteria but without *Blattabacterium* sequences ($n = 2,009,606$). If the sequence similarity of the bacterial database BLAST was $>2\%$ higher than the *Blattabacterium* database BLAST, then the sequence was removed. The threshold of 2% was chosen based on preliminary assessment of pairwise genetic divergence between *Blattabacterium* and close relative Bacteroidales.

Some of the inserts present in a host genome may be the result of a duplication or transposition of a previously inserted *Blattabacterium* sequence. To identify the number of unique HGT inserts within a genome (i.e., to count only those arising from separate transfers from a *Blattabacterium* genome), we applied a similar approach to that used for the false duplication filter above: inserts were extracted with 300 bp of flanking sequence, then BLASTed against all of the other inserts (with flanking) within the same genome. HGT inserts with a sequence identity >70% (alternative threshold >55%) and query coverage >70% (alternative threshold >55%) were considered to have been duplicated. Finally, manual auditing of the results was performed by additional BLASTs and manual inspections of the sequences themselves. We were particularly stringent when manually curating the termite putative inserts given the existence of termite gut symbionts that are closely related to *Blattabacterium* (e.g., *Bacteroides*/*Flavobacteria*). There was one potentially ancient *Blattabacterium* insert in the *Z. nevadensis* genome (Fig. 1A), but we were unable to confidently verify its authenticity with our methods.

The genomic features in which the filtered inserts are located (such as intergenic regions, exons, introns, promoters) were identified by comparing the positions of the inserts and genomic features using bedtools and custom AWK scripts. Genomic annotation files were those previously prepared using FGENESH++.

The minimum ages of ancestral inserts across the phylogeny were estimated by comparing sequence similarity of each insert and inserts in other cockroach species against the sequence similarity of each insert and the genome of the host's specific *Blattabacterium* strain. For species whose genomes have been sequenced multiple times by different research groups (i.e., *Periplaneta americana*, *Blattella germanica*, and *Zootermopsis nevadensis*), only the genome version with the most inserts was analyzed. Further, inserts associated from a duplication/transposition event, as characterized above, were excluded in this dating analysis. To identify putatively ancestral inserts, the filtered inserts were first extracted from the genomes using bedtools. Second, each insert from a given cockroach genome was BLASTed against the specific *Blattabacterium* strain that the species harbors (see Dataset S2). Only species with a published species-specific *Blattabacterium* strain were included ($n = 11$). Third, each insert was BLASTed against all of the inserts from the other cockroach/termite taxa (excluding itself). Fourth, the two BLAST output files were compared using a custom R script. The hit was set to 'NA' if query coverage was <70%.

If sequence identity was considerably higher for the host's specific *Blattabacterium* BLAST than for BLASTs of inserts from other cockroach/termite genomes, then the insertion was considered to have occurred subsequent to the host's divergence from the other species in the sample. In contrast, if sequence identity was higher for one or more inserts from the termite/cockroach library, than for the host's *Blattabacterium*, then the insert was considered ancestral to those species sharing it. Lastly, if sequence identities for the two BLAST hits were somewhat similar, we were unable to classify the age of the insert with confidence. Clear separations between these three cases were evident in each of the analyses (see the example given in aging analysis details on GitHub: https://github.com/MEEP-projects/HGT_pipeline/tree/main/other), and some manual auditing was carried out.

The divergence date of the most distantly related species both harboring the insert was treated as the minimum age of the insert. The resulting set of putatively ancestral, but unverified, inserts (see next paragraph about how ancestral inserts were validated) are presented in Dataset S5, which highlight that inserts are distributed across multiple evolutionary time points. To further investigate the temporal patterns of the HGT events, we used BLAST to map the sequences of all unique inserts against the species' specific *Blattabacterium* genomes (therefore, as above, this was restricted to the 11 cockroach species with published *Blattabacterium* strains; Dataset S2). Pairwise sequence identity between BLAST hits was used as a relatively coarse proxy for relative insert age, under the assumption that older inserts will have accumulated more sequence divergence (noting that substitution rates may not be uniform among inserts, although most

inserts were located in intergenic regions and may therefore experience relatively weak selective constraint). We then tested whether the distribution of pairwise identities conformed to a unimodal distribution using Hartigan's dip test [23] in the R package *dipTest* v.0.77-2 [24]. A significant deviation from unimodality was interpreted as evidence that inserts had originated via few, large HGT events that produced discrete peaks of pairwise identity. All distribution plots were also visually inspected to contextualise the results of the dip tests. Hits with 100% pairwise identity were excluded from this analysis, as these likely represent recent inserts yet to be subjected to significant purifying selection, and are therefore expected to follow a divergent frequency distribution. The results of these pairwise identities and dip tests are presented in Dataset S5.

We carried out additional analysis to verify the common origin of numerous inserts identified as putatively ancestral (i.e. putatively ancestral inserts from Dataset S5). We first identified all inserts that were found in two or more species. For each of these inserts, we then extracted the 5 kb upstream and downstream of the insert. Upstream flanking sequences were then assessed for sequence similarity using BLASTN. This was repeated for downstream flanking regions. Inserts whose flanking regions displayed a minimum sequence identity of 80% and an alignment length of at least 1000 bp were considered 'ancestral' (i.e. insertion occurred prior to divergence of the species with its sister taxon). A total of 3689 'ancestral' inserts of varying minimum inferred insertion times were identified. We confirmed the homology (and common origin) of 22 putatively ancestral inserts by inspecting 5 kb of cockroach genome sequence upstream and downstream from the insert position across multiple taxa (see Fig. 2A for an example). For all examined inserts, multiple taxa harbored homologous DNA across these ~10 kb regions, though some species were inferred to have subsequently lost the inserts.

We characterized the source locations of the inserts along the *Blattabacterium* genome. First, we aligned the inserts back to their respective *Blattabacterium* genomes using *bwa mem*. Only species with a published species-specific *Blattabacterium* strain were included in this analysis (Dataset S2). Note, not all of the inserts aligned, as some inserts were characterized based on fragments of other *Blattabacterium* genomes (not only the species-specific *Blattabacterium* strain). Second, we used a combination of *samtools* and *awk* to only retain primary alignments, and to record the insert start position along the alignment. Lastly, we plotted the cumulative start position counts along the *Blattabacterium* genome using the R packages *tidyverse* v2 [25] and *ggplot2* v3.5.1 [26], whereby a linear trend indicates that the origin of the inserts in the *Blattabacterium* genome is random.

We also assessed the distribution of inserts among and along the host genome contigs to investigate whether inserts were found in relatively few clusters, which would suggest few independent HGT events (which inserted large spans of *Blattabacterium* DNA subsequently broken apart by recombination), or whether they have a wide distribution across the genomes, suggesting many HGT events. First, we used a chi-squared goodness-of-fit test in R to determine whether the numbers of inserts per genomic contig differed significantly from the expected proportions under a purely random distribution (in which the proportion of inserts directly corresponds to the lengths of the contigs). To improve model fit, we limited this analysis to contigs with ≥ 2 inserts, and simulated *p*-values using a Monte Carlo test [27] with 10,000 replicates. In addition, to determine whether any significant results represented meaningful deviations from random distributions, we estimated effect size using Cramér's V in the R package *lsr* v.0.5.2 [28]. Most chi-squared tests showed significant deviation from neutral expectations ($p < 0.05$), however this result is not surprising considering the statistical power (i.e. number of contigs) in our analyses, and the deviations were found to be minimal based on effect sizes (Dataset S6).

Second, we used Kolmogorov–Smirnov (KS) tests to assess the randomness of the HGT insertion distribution within contigs. The KS tests assumed that if insertion position is random, then the distribution of insert positions over each contig will appear as a uniform distribution from position 1 to the end of the contig (minus 250 to account for the insert length). Contigs with fewer

than three inserts were excluded. The KS tests supported a random distribution of inserts within >85% of all contigs for every genome (Dataset S6). The distributions of inserts along the host contigs are depicted in Fig. 1C.

HGT insert QC

To robustly confirm that inserts were evident in raw genomic reads as well as the final assemblies, we aligned the raw reads against the respective assembled genomes. Raw reads that aligned only with the putative insert, or a subset thereof, were assessed to have likely derived from contaminant *Blattabacterium* DNA. In contrast, raw reads spanning all or some of the insert as well as 'native' cockroach flanking sequence were considered evidence that the insert was genuine. This step was undertaken on five genomes. See a schematic of this workflow and examples on GitHub (https://github.com/MEEP-projects/HGT_pipeline/tree/main/other).

We found that none of the identified inserts aligned with solely contaminant *Blattabacterium* reads; rather, each aligned with multiple raw reads spanning the boundary between exogenous inserted DNA and cockroach flanking sequence. In addition, in cases where *Blattabacterium* contaminant DNA was found to align to the insert (alongside genuine cockroach genomic reads), the *Blattabacterium* sequences were consistently divergent from the insert sequences.

To assess the sensitivity of HGT insert counts to our choice of parameters, we re-characterized inserts using alternative, less stringent thresholds (see the 'Characterizing HGT inserts' section). This led to a greater number of inserts being identified (246–5841 in the cockroach genomes), though overall patterns remained similar (Dataset S3). Note that the inserts characterized in the termite genomes using these thresholds were not manually audited, whereas the primary, more stringent dataset underwent particularly rigorous manual curation (given the presence of termite gut symbionts closely related to *Blattabacterium*).

To investigate whether the number of *Blattabacterium* inserts identified could reflect a methodological artefact, and to assess whether similar patterns are observed for other bacterial species, we repeated the HGT insert pipeline on *Buchnera* (an aphid symbiont) and *Wolbachia* (a bacterial genus known to infect some Blattodea species). The initial genome libraries used to characterize the HGT inserts comprised 37 and 20 *Buchnera* and *Wolbachia* genomes, respectively (Dataset S2).

We identified 0–21 putative *Buchnera*-derived inserts (average = 3.2) across the host genomes following the HGT pipeline. This number decreases when applying more stringent filtering parameters for excluding inserts that are similar to other bacterial species (see details in HGT pipeline above); i.e. most likely all putative *Buchnera*-derived inserts instead originate from *Blattabacterium* or other bacterial species. Indeed, manual inspection suggests that the majority of these hits originate from genomic regions that are relatively conserved across bacteria, such as 16S. This is consistent with the lack of co-occurrence between *Buchnera* and these hosts. We identified 1–35 *Wolbachia*-derived inserts (average = 9.5) per genome. The higher number relative to *Buchnera* likely reflects that *Wolbachia* do infect these hosts, although insert counts remain lower than those observed for *Blattabacterium*, potentially reflecting differences in host–bacterial interactions.

Since completing the analyses and original manuscript preparation, additional Blattodea genomes have become available. To assess whether the patterns identified in this study are consistent across these additional taxa, we applied the same HGT insert pipeline to five newly available genomes: one termite (*Mastotermes darwiniensis*; GCA_053541125.1), which is the only termite species known to harbour *Blattabacterium*; two *Cryptocercus* species (*C. punctulatus*; GCA_054643855.1 and *C. shennongjiansis*; GCA_054369535.1), which are a sister group to termites; and two additional cockroach species (*Ectobius pallidus*; GCA_964059185.1 and *Loboptera decipiens*; GCA_964662295.2).

Using the pipeline described above, we identified 48 inserts in *M. darwiniensis* (48 unique inserts; i.e. no genuine duplicates were removed), 69–173 inserts across the *Cryptocercus* species (68–173 unique inserts), and 33–98 inserts in the two additional cockroach species (33–94 unique inserts). These values are broadly consistent with those observed in the original dataset and align with expectations given the species' biology and phylogenetic position.

Dataset S1 (separate file). QC of newly assembled cockroach genomes.

Dataset S2 (separate file). Details of Blattodea, *Blattabacterium cuenoti*, *Buchnera aphidicola*, and *Wolbachia* genomes.

Dataset S3 (separate file). Number of HGT inserts within each genome, and the number of unique inserts (after excluding inserts likely arising from duplication or transposition) based on standard filtering (presented in manuscript, Fig. 1A) and alternative, less stringent thresholds.

Dataset S4 (separate file). Gene names for inserts found in exons based on homology predictions in FGENESH++.

Dataset S5 (separate file). Results of the preliminary ageing analyses (first tab), and a table of pairwise sequence identities between the HGT inserts and the host-specific *Blattabacterium* strains, alongside the results of the dip test (second tab).

Dataset S6 (separate file). Results of the Chi-Square test analyses (and associated effect sizes) and Kolmogorov–Smirnov (KS) test analyses for assessing the distribution of inserts among and along the host genome contigs.

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